

V BOOPATHY KHANISHK

Set 4: Product Inventory Management

Dataset: Product Inventory

Product ID Product Name Quantity Available

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title

the chart.

2. Generate a stacked bar chart to show the quantity of each product within different product categories.

3. Build a scatter plot to explore the relationship between product price and quantity available. Explain

the findings.

4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact

with the data.

SOLUTION :

CODE :

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as gg
from plotly.subplots import make_subplots
```

```
df_inventory = pd.DataFrame({
    'Product_ID': [1, 2, 3, 4, 5],
    'Product_Name': ['Product A', 'Product B', 'Product C', 'Product D', 'Product E'],
    'Category': ['Electronics', 'Electronics', 'Apparel', 'Apparel', 'Home Goods'],
    'Quantity_Available': [250, 175, 300, 200, 220],
    'Price': [120, 250, 45, 60, 150]
})
```

```
plt.figure(figsize=(8, 4))

plt.bar(df_inventory['Product_Name'], df_inventory['Quantity_Available'],
color='mediumseagreen', edgecolor='black')

plt.title('Product Inventory Quantities', fontsize=14)

plt.xlabel('Product Name', fontsize=12)

plt.ylabel('Quantity Available', fontsize=12)

for i, val in enumerate(df_inventory['Quantity_Available']):
    plt.text(i, val + 5, str(val), ha='center', fontweight='bold')

plt.tight_layout()

plt.show()
```

```
cat_pivot = df_inventory.pivot(index='Category', columns='Product_Name',
values='Quantity_Available').fillna(0)

cat_pivot.plot(kind='bar', stacked=True, figsize=(8, 4), colormap='Set2')

plt.title('Product Quantity Distribution across Categories', fontsize=14)

plt.xlabel('Category', fontsize=12)

plt.ylabel('Quantity Available', fontsize=12)

plt.xticks(rotation=0)

plt.legend(title='Product')

plt.tight_layout()

plt.show()
```

```
plt.figure(figsize=(8, 4))

sns.scatterplot(data=df_inventory, x='Price', y='Quantity_Available', hue='Product_Name',
s=120)

plt.title('Product Price vs. Quantity Available', fontsize=14)

plt.xlabel('Price ($)', fontsize=12)

plt.ylabel('Quantity Available', fontsize=12)

plt.grid(True, linestyle='--', alpha=0.6)

plt.tight_layout()

plt.show()
```

```
print("--- Insights from Scatter Plot ---")
```

```
print("Higher-priced products (e.g., Product B at $250) tend to have lower inventory stock levels (175 units),")
```

```
print("whereas lower-priced products (e.g., Product C at $45) maintain higher stock levels (300 units) to meet expected volume demand.\n")
```

```
fig = make_subplots(
    rows=1, cols=2,
    subplot_titles=("Inventory Level per Product", "Price vs. Quantity")
)
```

```
fig.add_trace(gg.Bar(x=df_inventory['Product_Name'],
y=df_inventory['Quantity_Available'], marker_color='forestgreen'), row=1, col=1)
```

```
fig.add_trace(gg.Scatter(x=df_inventory['Price'], y=df_inventory['Quantity_Available'],
mode='markers+text', text=df_inventory['Product_Name'], textposition='top center',
marker=dict(size=12)), row=1, col=2)
```

```
fig.update_layout(title_text="Set 4: Inventory Management Dashboard", showlegend=False,
height=450)
```

```
fig.show()
```

OUTPUT :



